

CSA12 MCQ

Total points 28/30 ?

Register Number *

192425117

Name *

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✓ A processor has registers R1, R2, and R3. During one clock cycle, the control unit performs $R3 \leftarrow R1$ while the ALU simultaneously computes $R2 + R4$. What is the main advantage of this operation? *1/1

- ☐ A. It converts all operations into memory operations
- ☐ B. It prevents data movement between registers
- ☐ C. It eliminates the need for registers
- ☒ D. It allows independent operations to occur concurrently ✓

✓ A processor executes the micro-operation $R2 \leftarrow R1 + R3$. Which sequence correctly represents the required data movement? *1/1

- ☒ A. $R1 \text{ and } R3 \rightarrow \text{ALU} \rightarrow R2$ ✓
- ☐ B. $R1 \rightarrow R2 \rightarrow R3 \rightarrow \text{ALU}$
- ☐ C. $R2 \rightarrow \text{ALU} \rightarrow R1 \text{ and } R3$
- ☐ D. $\text{ALU} \rightarrow R1 \text{ and } R3 \rightarrow R2$



✓ During debugging, a student observes that R4 contains the result of R1 AND R2, while the original contents of R1 and R2 remain unchanged. Which architectural feature explains this behavior? *1/1

- ☐ A. The control unit converts AND into a transfer operation
- ☐ B. The memory unit automatically restores source registers
- ☒ C. Source registers provide operands while the destination register receives the result ✓
- ☐ D. ALU operations normally overwrite all source registers

✓ A processor needs to calculate $R5 \leftarrow R5 + 1$ repeatedly for a loop counter. Which ALU operation is most directly suited to this task? *1/1

- ☐ A. XOR
- ☒ B. Increment ✓
- ☐ C. Complement
- ☐ D. AND

✓ A register contains 10110110 and another contains 11001100. The ALU performs a bitwise AND. What should the destination register contain? *1/1

- ☐ A. 01000010
- ☐ B. 01111010
- ☐ C. 11111110
- ☒ D. 10000100 ✓



✗ A control unit performs the following sequence: $R1 \leftarrow R2$, followed by $R3 \leftarrow R1$. If both operations occur in separate clock cycles, what value will R3 receive? *0/1

- ☐ A. The original value of R2
- ☐ B. The address of R2
- ☒ C. The original value of R1
- ☐ D. The ALU status flags only

Correct answer

- ☒ A. The original value of R2

✓ A processor executes $R4 \leftarrow R1 - R2$ using the ALU. Which additional information is particularly useful for determining whether an unsigned borrow occurred? *1/1

- ☐ A. Memory address register
- ☐ B. Program counter value
- ☒ C. Carry/borrow status
- ☐ D. Instruction opcode only



✓ An embedded controller needs to clear selected bits in a control register *1/1 without modifying other bits. Which operation is most appropriate?

- ☐ A. Increment operation
- ☐ B. OR with a suitable mask
- ☐ C. Addition with a suitable mask
- ☒ D. AND with a suitable mask ✓

✓ A register contains 11001101, and the processor performs a bitwise OR *1/1 with 00110010. What is the resulting value?

- ☒ A. 11111111 ✓
- ☐ B. 00000000
- ☐ C. 11001110
- ☐ D. 00110010

✓ A processor performs several register-transfer operations, but the destination register occasionally receives an incorrect value because two sources attempt to drive the same internal data path simultaneously. Which problem is most directly indicated? *1/1

- ☐ A. Pipeline starvation
- ☒ B. Bus contention ✓
- ☐ C. Branch misprediction
- ☐ D. Register overflow



✓ A processor uses three internal buses so that two registers can supply operands to the ALU while another register receives the result in the same cycle. What is the primary benefit? *1/1

- ☐ A. Removal of memory latency
- ☐ B. Elimination of the ALU
- ☒ C. Increased parallel data transfer ✓
- ☐ D. Reduced instruction-set size

✓ A single-bus processor requires three clock cycles to perform $R3 \leftarrow R1 + R2$: one for transferring R1, one for transferring R2, and one for storing the result. A three-bus design performs the two operand transfers and result transfer within one cycle. What is the ideal reduction in transfer cycles? *1/1

- ☐ A. From 2 to 1
- ☐ B. From 3 to 2
- ☐ C. From 1 to 3
- ☒ D. From 3 to 1 ✓



✓ A processor has separate buses for reading two operands and another path for writing the ALU result. Which operation benefits most directly from this organization? *1/1

- ☒ A. $R5 \leftarrow R1 + R2$ ✓
- ☐ B. Resetting the processor
- ☐ C. Reading one instruction byte from memory
- ☐ D. Clearing the program counter

✓ A multi-bus system experiences delays because two functional units repeatedly request the same bus at the same time. Which factor is primarily responsible? *1/1

- ☐ A. Instruction decoding
- ☐ B. Register overflow
- ☒ C. Bus contention ✓
- ☐ D. Sign extension



✓ A processor changes from a single internal bus to a multiple-bus organization but observes little performance improvement for a workload dominated by memory access delays. What is the most reasonable explanation? *1/1

- ☐ A. Multiple buses always reduce clock frequency
- ☒ B. Memory latency has become the dominant bottleneck ✓
- ☐ C. Registers cannot operate with multiple buses
- ☐ D. Multiple buses cannot improve ALU operations

✓ During execution of $R4 \leftarrow R1 + R2$, Bus A carries R1 and Bus B carries R2 *1/1 to the ALU. Which component determines the arithmetic operation?

- ☐ A. Program counter
- ☐ B. Memory unit
- ☐ C. I/O interface
- ☒ D. ALU control logic ✓



✓ A processor has two source buses and one destination bus. An instruction requires three independent source operands simultaneously. What is the most likely limitation? *1/1

- ☐ A. The destination bus becomes an input bus
- ☐ B. Memory automatically supplies the third operand
- ☒ C. Only two operands can be supplied directly through the available source buses ✓
- ☐ D. The ALU cannot perform arithmetic

✓ A compiler schedules instructions so that independent register transfers use different buses simultaneously. Which architectural characteristic is being exploited? *1/1

- ☒ A. Parallel bus utilization ✓
- ☐ B. Cache replacement
- ☐ C. Speculative execution
- ☐ D. Branch prediction



✓ A processor uses five stages: Fetch, Decode, Execute, Memory, and Write-back. *1/1
Instruction I1 enters Fetch during cycle 1. In which cycle does I1 normally complete Write-back if each stage requires one cycle and there are no stalls?

☐ A. Cycle 4

☐ B. Cycle 6

☐ C. Cycle 3

☒ D. Cycle 5 ✓

✓ A five-stage pipeline processes 10 independent instructions without stalls. Assuming one cycle per stage, how many cycles are required? *1/1

☐ A. 15

☐ B. 50

☒ C. 14 ✓

☐ D. 10



✓ A non-pipelined processor requires five cycles per instruction, while a five-stage pipeline requires approximately one cycle per instruction after filling. For a very large number of independent instructions, what is the theoretical maximum speedup? *1/1

- ☒ A. 5 ✓
- ☐ B. 10
- ☐ C. 3
- ☐ D. 2

✓ A pipeline produces one completed instruction every cycle after its initial fill period. Which performance measure is primarily improved? *1/1

- ☐ A. Memory capacity
- ☒ B. Instruction throughput ✓
- ☐ C. Instruction size
- ☐ D. Register width

✓ A four-stage pipeline has stage delays of 2 ns, 3 ns, 5 ns, and 2 ns. Ignoring pipeline-register overhead, what is the minimum clock period? *1/1

- ☐ A. 12 ns
- ☐ B. 3 ns
- ☒ C. 5 ns ✓
- ☐ D. 2 ns



✓ A five-stage pipeline has a clock period of 4 ns. Pipeline-register overhead increases the required period by 1 ns. What is the actual clock period? *1/1

- ☒ A. 5 ns ✓
- ☐ B. 20 ns
- ☐ C. 4 ns
- ☐ D. 3 ns

✓ A pipeline executes 20 instructions in 24 cycles, whereas the corresponding non-pipelined processor requires 100 cycles. What is the achieved speedup? *1/1

- ☐ A. 20.0
- ☐ B. 5.0
- ☐ C. 2.4
- ☒ D. 4.17 ✓

✓ A pipeline is full, but one stage takes twice as long as every other stage. *1/1
What is the most direct consequence?

- ☐ A. The pipeline automatically skips the slow stage
- ☒ B. The slowest stage determines the pipeline clock period ✓
- ☐ C. The fastest stage determines the clock period
- ☐ D. All instructions execute in one cycle



✓ A processor increases the number of pipeline stages from 5 to 10, but the clock frequency does not increase because pipeline-register overhead becomes significant. What does this illustrate? *1/1

- ☐ A. Throughput becomes independent of clock frequency
- ☐ B. Pipeline hazards disappear with more stages
- ☒ C. Pipeline depth has practical limitations ✓
- ☐ D. Deeper pipelines always provide linear speedup

✗ A six-stage pipeline completes its first instruction after six cycles and then completes one instruction per cycle. If 15 instructions are executed without stalls, how many cycles are required? *0/1

- ☒ A. 21 ✗
- ☐ B. 15
- ☐ C. 90
- ☐ D. 20

Correct answer

- ☒ D. 20



✓ Consider the instructions: I1: $R1 \leftarrow R2 + R3$ and I2: $R4 \leftarrow R1 + R5$. I2 needs the value produced by I1. Which hazard occurs? *1/1

- ☐ A. WAW hazard
- ☒ B. RAW data hazard ✓
- ☐ C. Structural hazard
- ☐ D. WAR hazard

✓ Two pipeline instructions require the same single memory port during the same cycle. Which hazard is this? *1/1

- ☒ A. Structural hazard ✓
- ☐ B. Register hazard
- ☐ C. Control hazard
- ☐ D. Data hazard

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